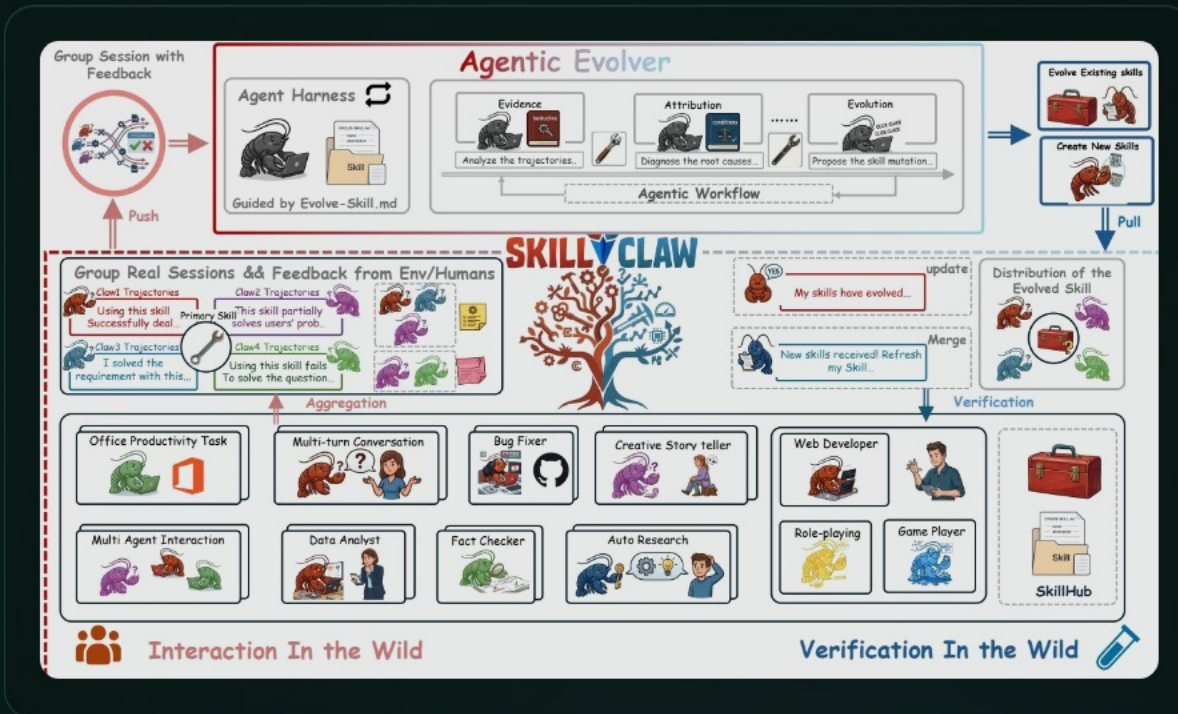


ARXIV.ORG/ABS/2604.08377

SkillClaw: Collective Skill Evolution

LLM agent skills continuously evolve by aggregating cross-user interaction trajectories and applying agentic, evidence-driven updates.

Multi-user agent ecosystems · agentic evolver



PROBLEM FRAMING

Static skills make every user rediscover the same fixes.

CURRENT PARADIGM

Manual install, no runtime evolution

- Users install reusable skills from a centralized hub.
- Skills remain static after deployment.
- Failures and recoveries rarely persist beyond a session.



SKILLCLAW TARGET

System-level knowledge accumulation

- Repeated patterns become evidence for evolution.
- Shared skill libraries improve from usage.
- Updates synchronize back to all participating agents.

Closed loop: multi-user interaction to shared skill evolution.

Evidence

Structured trajectories preserve action-feedback causal chains.

Attribution

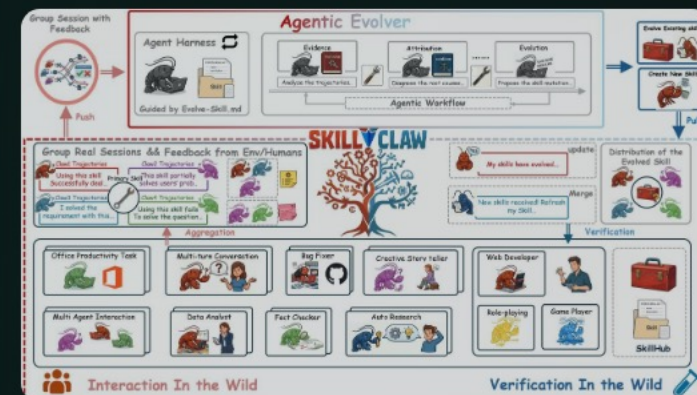
Skill-specific groups expose root causes of failures and successes.

Evolution

Agentic reasoning proposes skill mutations or new skills.

Distribution

Verified updates merge into the repository and synchronize to agents.



Pipeline from multi-user interaction → session collection → skill evolution → skill synchronization.

AGENTIC EVOLVER PIPELINE

Open-ended reasoning over evidence, not predefined rules.

01

Evidence

Analyze skill-specific trajectories to identify behavioral patterns in full action-feedback chains.

02

Attribution

Diagnose root causes behind failures and successes across grouped sessions.

03

Evolution

Propose mutations that refine existing skills or create new ones, then verify before merge.

CROSS-USER AGGREGATION

Diverse contexts reveal a skill's behavioral boundary.

Trajectories from compatible Claw-style systems exercising the same skill produce complementary signals. A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

OpenClaw

CoPaw

IronClaw

PicoClaw

ZeroClaw

NanoClaw

NemoClaw

Shared evidence base

Success patterns

Conditions under which a skill works.

Skill grouping

Sessions aggregate by referenced skills.

Failure modes

Contexts where behavior breaks.

Evolution signal

Evidence supports generalizable updates.

CASE STUDY • MULTIMODAL CREATIVE PIPELINE

Conservative verification: only one candidate enters the pool.

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accepted candidates

DATE	OUTCOME	CANDIDATE	RATIONALE
Day 1	Accepted	validate-tmp-workspace-inputs	Check /tmp_workspace inputs and environment setup; upgraded to new best pool.
Day 2	Rejected	multimodal-input-validation-and-setup	Multimodal input validation and output environment init; current best pool kept.
Day 3	Rejected	multimodal-creative-task-pipeline	Extract from PDF/video/image; current best pool kept.
Day 4	Rejected	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation; current best pool kept.
Day 5	Rejected	creative pipeline + validate-required-input-files	Per-file fail-fast validation; current best pool kept.
Day 6	Rejected	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths; not admitted to next cycle.

CASE STUDY • GIT PUSH AUTH FALLBACK

Iterative refinement succeeds until evidence plateaus.

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accepted updates

DATE	OUTCOME	CANDIDATE	RATIONALE
Day 1	Accepted	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure; added to Safety best pool.
Day 2	Accepted	git-push-with-auth-fallback	Unified patch/bundle filenames and reduced inconsistency; updated best pool kept.
Day 3	Accepted	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths and secrets audit; fixed subshell pitfalls.
Day 4	Current pool	(none)	Same-pool retest; validator confirmed current pool as best and continued it.
Day 5	Rejected	git-push-with-auth-fallback	Added “push hang treated as auth failure”; no improvement.
Day 6	Rejected	git-push-with-auth-fallback	Added identity config and filename consistency; not admitted to next cycle.

Three properties distinguish SkillClaw from static skill systems.

I

Collective evolution

Individual interactions contribute to a shared, continuously improving skill ecosystem.

II

Fully automatic

Evolution is driven by runtime interaction without manual curation or explicit user intervention.

III

Agentic paradigm

Open-ended reasoning over evidence produces context-aware skill mutations.

Implication: continuously improving agent systems can convert distributed runtime experience into verified shared capabilities, rather than leaving every user to solve the same failures alone.